Speech in the Panaudikon: On the Relationship between Speech Processing Research and Surveillance

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Abstract

The potential of using speech technologies for surveillance purposes are tacitly acknowledged within the field of speech processing, however there has been little research on how theories from surveillance studies interact with speech processing research. In this paper, we analyse 20% of papers published in 2025 from one of the main speech venues, Interspeech, through Steve Mann's compass of veillances. We find that engagement with considerations of surveillance is almost entirely absent, despite risks being readily apparent in a number of research tasks, with some tasks being more prone to risks of surveillance. Encouragingly, we find tasks and findings which provide avenues for counter-surveillance, yet these introduce other risks. We argue that identifying mechanisms for addressing the risk of surveillance requires discussion, and therefore provide a framework for researchers to engage in to identify how their work engages with surveillance, and how they can limit the risks of their research. To this end, we present a list of questions for researchers to engage in how their work might contribute to surveillance.

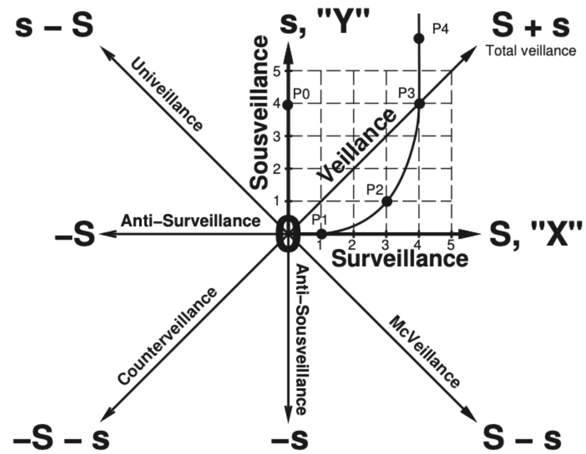

Figure 1: Steve Mann's 8-point Veillance Compass (Mann 2013). Reproduced with license.

1 Introduction

Surveillance is a factor that touches upon many of our daily lives, from being recorded by closed-circuit television (CCTV) in public spaces (Mann 2016) with and without facial recognition technologies, to the content we post on social media having consequences for our ability to obtain visas (Brown 2014), or merely how our online footprint is tracked: surveillance has come to touch upon most of our public lives. Yet, statements that we have voiced have been less subject to widespread surveillance, in part due to the difficulties in processing speech in the wild (Zhao et al. 2025). In fact, when Amazon made headlines in 2021 for its Rekognition system being deployed for transcribing phone calls from prisons in the United States of America, the company's spokespeople acknowledged that "heavy accents or poor audio quality can lead to variations" (Sherfinski and Asher-Schapiro 2021). Recently, speech processing technologies have seen rapid improvement, and deployment. From speech-based assistant systems such as Amazon Alexa and Apple Siri, to speech transcription services included in video-calling software, to speech synthesis systems such as those used in large language models, speech technologies are increasingly being embedded into our society.

Historically, surveillance has required painstaking manual labour, the increased capabilities of machine learning systems provide an avenue for improved data processing, and thereby surveillance, mechanisms. While the prospect of increased surveillance capabilities may raise concerns for many, for some, it is a "vision" to be fulfilled. Indeed, in January 2026, the United Kingdom's Home Secretary, Shabana Mahmood, stated that:

> "[M]y ultimate vision for that part of the criminal justice system was to achieve, by means of AI and technology, what Jeremy Bentham tried to do with his Panopticon. That is that the eyes of the state can be on you at all times [...] I think there's big space here for being able to harness the power of AI and tech to get ahead of the criminals" – *Shabana Mahmood* (Morrison 2026).

While risks of dual use of speech technologies have been explicitly discussed around synthesis (Hutiri, Papakyriakopoulos, and Xiang 2024; Burgess, Carlon, and Doyuran 2025), research around surveillance has centred questions of privacy-preservation (Tomashenko et al. 2026) and little attention has been given to how research areas in speech processing engage with surveillance—where privacy is centred around personal data, while surveillance extends to include the content of ones speech. This is concerning because (i) even where text can more easily be processed, speech signals carry information about the speaker, their emotional state, sex, and other characteristics (Al-Ali et al. 2025); (ii) it is unclear which research areas of speech processing elicit risks of surveillance, which risks are surfaced, and how to address them; and (iii) the lack of a holistic overview of the risks of the field act as hindrances to organise against being subject to surveillance, such as in the case of the successful campaigns against facial recognition which has resulted in restrictions in their use in 16 states, and outright bans in 15 cities, in the United States of America (State of Surveillance 2025).

In this paper, we seek to close this gap by analysing speech processing papers for how they engage with questions of dual use and surveillance. Drawing on a theory of *veillance*, we annotate 229 (20%) of papers published at Interspeech 2025. Discouragingly, we find a dearth engagement with dual use and risks of surveillance. Specifically, only three papers mention dual use and only two papers mention surveillance in our sample. Still, highly encouragingly, we find several areas of speech research that affords counters to surveillance, including text-to-speech, voice conversion, and privacy-preserving methods such as speaker anonymisation. Further, we provide an in-depth discussion around the nuances of surveillance. Although questions of surveillance are often presented as clear-cut, the potential for speech synthesis technologies complicates this view: on one hand, synthesis may provide additional mechanisms for privacy in the context of surveillance, yet the very same tools may also further perpetuate harm to individuals, e.g. by providing mechanisms for scammers to remain undetected. We argue that rather than seeking to limit, prohibit, or encourage certain areas of research, a more appropriate method would be to encourage discussion around how work might engage with surveillance, and provide guidance for mechanisms towards this. Finally, the paper provides concrete recommendations for the field and a set of questions for researchers to interrogate how their work engages with surveillance and how risks might be mitigated.

Paper Structure In the remainder of this paper, we provide an overview of work on surveillance in Section 2. In Section 3 present our data collection and annotation methodology. We then present the outcomes of our analyses in Section 4, and a discussion on the nuances of surveillance in speech processing research in Section 5. We close the paper with recommendations for the field and questions to help researchers reflect on the surveillance capacity of their work in Section 6.

2 Background

2.1 Compass of Veillances

Surveillance practices have long existed as a way to exercise control over marginalised groups, from the Roman Empire's provincial information collectors, the Frumentarii (Sinnigen 1962), to the enslaved people during the 18th and 19th century United States of America (Browne 2015) to modern day surveillance enabled through CCTV (Mann 2016) and the internet (Fuchs et al. 2013). In (Browne 2015), Browne argues that not everyone is equally subject to surveillance. Drawing a history from the transatlantic slave trade to modern biometrics, she argues that marginalised people, and Black people in the United States in particular, have been subject to an othering through technological means—first through the distribution of news and wanted ads in which enslaved people who had escaped were described as property bearing distinct marks, such as scars to identify them; then through lantern laws which sought to make hyper-visible the Black people who were out after dark, and limited to three people per lantern; and now through modern means of biometrics. Rather than function purely as mechanism for being watched, surveillance of Black people and marginalised communities have served as mechanisms of control, to limit their movement and their ability to organise (Browne 2015). Yet, have other forms of looking and exercising of control occurred? During the French Revolution, citizens would routinely attend discussions of the national convention in an effort to exert power over the legislature (Maslan 1995). Similarly, in modern times, mobile phone recordings of law enforcement officers functioned as mechanisms to mitigate unlawful conduct by the officers (Andrejevic 2011). From this it is clear that (i) being watched is to have some form of force being applied to an individual and (ii) watching can be done from a position of power, e.g. through CCTV, or as a means of holding power to account (Mann 2016).

In his consideration of computer vision and its potentials for misuse, Steve Mann developed his compass of veillances (Mann 2013). In this theory he outlines eight axes of the act of seeing, arguing that each form of axes gives rise to different politics and different relationships between actors (see Figure 1): `Surveillance` refers to a third-party watching "from above" (Mann 2016), typically referring to the state or an actor in a position of power. Surveillance functions as a way for the state to enact power, and to limit the ability of the surveilled to act freely. Notably, in Bentham's (Bentham 2010) writings on the Panopticon, he noted that the purpose of the design of the prison was to instil the experience of being watched as a mechanism for ensuring that prisoners did not break the rules of the prison at any point. In contrast to surveillance, `Sousveillance` refers to a public watching "from below", referring to practices such as citizen's recordings. However, as Jansen (Andrejevic 2011) argues, for sousveillance to be effective, there must be some expectation of accountability and a realisation thereof. Without either, any recording of the agents in power is rendered meaningless in ensuring they act in concordance with the laws or rules they are subject to. `Univeillance` refers to the ability of a first-party in a communicative context to engage in a recording of another first-party. For example, in an interaction between parties that are equal amongst each other, either party can engage in univeillance by recording the other part (regardless of the other's knowledge or consent) . `Veillance`, in Mann's theory, refers to the general act of monitoring and encompasses sur-, uni- and sousveillance. In opposition to monitoring, the compass contains `Anti-surveillance`, which refers to methods by which one limits the ability of surveillance to function, either by disabling the surveillance mechanisms themselves or by rendering oneself invisible to them. `Anti-sousveillance` on the other hand functions as a mechanism to prevent sousveillance from occurring. When this is instituted as policy, as in airport passport security lines where individuals are not allowed to record images or photos while simultaneously being under the surveillance of airport cameras, it becomes `McVeillance`, whereas it is mediated in other ways, such as by disabling the recording devices that are engaging in sousveillance or erecting a barrier that prevents recording. `Counterveillance` refers simply to the ability to prevent being watched or recorded, and covers anti-surveillance, anti-sousveillance, and McVeillance.

2.2 Affordance Theory

In order to assess what particular form of veillance a given paper gives rise systematically, we turn towards the concept of *affordances*. The concept was first described by Gibson (Gibson 1983) and seeks to describe how humans are able to sense how to engage and interact with our environment in such a way that it is conducive to our goals and purposes. The core idea behind the theory is the notion that some objects afford certain outcomes in specific environments. For example, a small pebble near a lake has the affordance of being thrown across the surface of the lake. A large rock, however, does not have the affordance of being possible to skip across the surface of a lake. Similarly, the same small pebble may function as a poor tool breaking other stones, whereas the larger rock, with its larger surface area, is a better candidate for breaking stones. That is, where the small rock affords skipping, in the context of a human by a still lake, the larger rock affords being used by a tool-making animal as a tool for breaking other rocks, as a function of its size. More precisely, the affordance of an object refers to the action potential of the object, i.e., what an individual with a particular purpose can do with the object.

In terms of speech technologies and their capacity for surveillance, we can think of the affordance of a research object as what a person can do with the object without significant technological reconfiguration (Majchrzak 2013). For example, an ASR system developed for language **A** may be possible to deploy for language **B**, however the transcripts that would arise from it are likely to be riddled with errors. We can therefore say that our ASR model trained for language **A** affords transcription for language **A** more so than it affords transcription for language **B**.

In terms of Mann's axes of veillance, we ask for each paper which axes it affords, and how well it affords that axis. For example, a lightweight ASR system such as the model proposed by Kong, Huang, and Ou (2025) affords sousveillance and univeillance more than it affords surveillance, because its compute requirements are low enough that it can be developed and used with minimal resources, thereby allowing a technically-minded person the ability to set up the system, develop it, and run it locally to process whatever speech they may wish to. Sousveillance and univeillance may often have overlapping affordances, but are distinct along axes of power. For example, while an ASR system for children's speech affords surveillance–by the hospital recording–and univeillance–through recording from another figure of authority–yet does not afford sousveillance as recording does not place children in a position of power.

2.3 Speech (and Surveillance)

To the best of our knowledge, the multifaceted relationship between speech technologies and surveillance has not been explicitly discussed in the literature. Some works have broadly explored how audio processing may enhance different surveillance tasks (Crocco et al. 2014), without addressing the potential negative impacts of such contributions. The impact of the growing amount of online speech data on surveillance has been analysed from a sociological perspective (Caroleo 2025). There is a lack of work on the negative contributions of how speech technologies advance surveillance; however, some papers address privacy in speech technology (Bäckström 2025), where surveillance is considered as a risk associated with privacy breaches.

2.4 Machine Learning and Surveillance

Surveillance spans across several data modalities, most notably text, audio, and video. Most modern speech processing, computer vision, and NLP research utilises machine learning models. The rise of mass surveillance is tightly connected to the scientific progress in computer vision (Kalluri et al. 2025). The performance of machine learning models, particularly deep neural networks, relies heavily on large amounts of data and substantial computational power. Institutions that practise surveillance are able to collect and store large amounts of data and have means to own powerful GPUs for computing. Larger datasets improve the performance of models used for surveillance, which in turn allows more precise surveillance. The ethical implications of large-scale dataset collection were discussed in speech processing (speaker recognition task) (Leschanowsky et al. 2025) and computer vision (Prabhu and Birhane 2020).

3 Methodology

3.1 Data Collection

We collected our initial sample of all $1,147$ objects retrieved using the SemanticScholar Search API, published at Interspeech 2025.[1] This sample includes papers and four keynotes. Removing the keynotes, we are left with $1,143$ papers as our initial sample. We select papers published at Interspeech for two reasons. First, Interspeech is a premier venue for speech technology and processing research, and work published covers a breadth from linguistically focused research (e.g., Oh, Kim, and Chung 2025), to more signal processing oriented work (e.g., Kim and Chung 2025), and to system demonstrations (e.g., Dewhurst et al. 2025). Second, the themes of Interspeech over the past few years have

[1]The collection of papers was conducted on the 5th of January, 2026.

Annotation category	Type	Options (if categorical)
Author Affiliation	Free Text	
Author Affiliation Type	Categorical	Academia, Governmental, Industry, Research Institute
Languages Included	Free Text	
Tasks	Free Text	
Task Category	Multi-label	ASR, Speech translation, Speaker identification, Speech synthesis, Speech enhancement, Anti-spoofing, Emotions or Traits, Clinical settings, Multimodal speech decoding, Speech representation, Dialogue systems, Environmental monitoring, Research analysis, Other
Datasets Used	Free text	
Use of Biomarkers	Multi-lable	Models biomarkers, Predicts biomarkers, Other, N/A
Open-Sourcing	Categorical	Open-source, Closed source, Restricted access, Not mentioned, N/A
Dual Use	Categorical	Discusses dual use, Mentions dual use, Not mentioned, N/A
Surveillance	Categorical	Discusses surveillance, Mentions surveillance, Not mentioned, N/A
Veillance Compass Areas	Multi-label	Surveillance, Anti-surveillance, Sousveillance, Anti-sousveillance, Univeillance, N/A
Other Risks	Free text	
Limitations	Categorical	Discusses limitations, Mentions limitations, Not mentioned, N/A
Other Funding Source	Free text	

Table 1: Overview of annotation schema.

Course-grained Category	Fine-grained Category	% Agreement
Veillance Axes	anti-sousveillance	91.43%
	anti-surveillance	85.71%
	sousveillance	68.57%
	surveillance	71.43%
	univeillance	68.57%
	N/A	85.71%
Biomarkers	Models biomarkers	85.71%
	Predicts biomarkers	91.43%
	Biomarkers: Other	97.14%
	N/A	80.00%
	Open Source	85.71%
	Dual Use	100%
	Discusses Surveillance	100%
	Risks	91.43%
	Limitations	77.14%

Table 2: Annotator agreement.

centred inclusion and ethics, thus encouraging work to address questions of speech technologies' social impact. From our initial sample of $1,143$ papers, we select 20 for validating our annotation schema, and 229 (20%) for annotation.[2]

3.2 Data Annotation

Two of the authors engaged in the annotation process, each annotating an independent sample from one another. To ensure consistency in annotation, we compute annotator agreement on 35 samples that were jointly annotated by both annotators, and conducted regular check-ins between the annotators to ensure that any questions and differences in understanding were identified and resolved. We annotated papers along the 12 dimensions:

[2]We ensure that there is no overlap our schema validation sample and our annotation sample.

Author Affiliation: We recorded the author affiliations in a first pass, and conduct a second pass over papers to group affiliations into academia, industry, government, and research institutes.
Languages Covered: Where research deals with speech signals, rather than room noise, echo, and other environmental factors, we record the languages that the research covers.
Tasks: We here conduct a fine-grained annotation of the task papers are conducting (e.g., automatic speech recognition (ASR) for dysarthric speech) on the basis of the description in the work. Once we had concluded the annotation, we conducted a second pass over tasks, to group them into one or more coarse grained categories through an inductive coding.[3]
Datasets Used: We record the datasets that are included for either training, evaluation, or analysis.
Use of Biomarkers: We note if biomarkers are used in the study, with the following options: "`models biomarkers`", "`predicts biomarkers`", "`other`", and "`N/A`".
Artefact Release: We collect information about how research artefacts are distributed by what is mentioned in research papers and categorise into the following categories: "`open-source`", "`closed source`", "`restricted access`", "`not mentioned`", and "`N/A`". We choose what is explicated in the papers themselves, i.e., papers are labelled as `open-source` if the authors provide a link to a website or repository holding the research artefacts, or explicitly mention that it is open source.
Dual Use Considerations: We select one or more categorical labels for dual use considerations from: "`Discusses`

[3]We identify the following task categories: *ASR, Speech Translation, Speaker Identification, Speech Synthesis, Speech Enhancement, Anti-Spoofing, Emotions or Traits, Clinical Settings, Multimodal Speech Decoding, Speech Representation, Dialogue Systems, Environmental Monitoring, Research Analysis*, and *Other*.

Dual Use", i.e., instances where concrete dual use concerns are raised and discussed; "Mentions Dual Use" which covers instances where concrete or abstract mentions of dual use without further discussion, e.g. "However, these techniques also pose significant risks, particularly in the spread of highly realistic misinformation" (Kim et al. 2025); "No Mention" where there is no mention of dual use; or "N/A" where dual use considerations are not relevant, e.g., in papers that perform evaluation of pre-existing methods.

Surveillance Considerations: We also categorically label how surveillance is treated in papers, allowing selection between: "Discusses Surveillance"; "Mentions Surveillance"; "No Mention"; or "N/A".

Mann's Veillance Compass Axes: We label each paper according to Veillance Compass Axes from Mann (2013), with the exception of Veillance, because in practice it exclusively co-occurs with sur- and sousveillance; Counterveillance similarly exclusively occurs along instances of anti-sousveillance and anti-surveillance; and McVeillance, as technologies by themselves do not set policies that allow some to record while limiting others. The final set of axes that are labelled for are: surveillance, anti-surveillance, sousveillance, anti-sousveillance, univeillance, and N/A. We label each paper according to the compass areas by drawing on affordance theory (Gibson 1983). This approach means that two papers for ASR may be labelled differently. For example, a paper which seeks to develop high performing ASR may be labelled as surveillance and not sousveillance if it requires a large number of GPUs to train and/or run. A paper may be labelled for multiple axes of veillance, for instance where a paper that affords anti-surveillance also affords anti-sousveillance.

Further, we ignore the question of temporality, when the processing of recorded speech can occur at a different point in time than the audio is recorded.

Other Risks While surveillance and dual use are a particular set of risks, we here capture other risks of harms that are raised by research papers. We categorise papers into "Discusses Risks"; "Mentions Risks"; "No Mention"; or "N/A".

Limitations All research methods have their limitations, some of which may relate to surveillance or the potential for the work to engage in surveillance. We therefore extract limitations according to the categories "Discusses Limitations"; "Mentions Limitations"; or "No Mention".

Funding Source: We extract mentions of different funding sources that are listed in the acknowledgements.

Where dual use, surveillance, limitations, and other risks are at least mentioned, we also extract the relevant segments of the papers (for an overview of our categories, see Table 1).

Annotator Agreement We randomly sample 35 papers for mutual annotation between the two annotators, and compute percentage-wise agreement on all categorical and multi-label dimensions (see Table 2), with the exception author affiliation type, and task category. We exclude these,

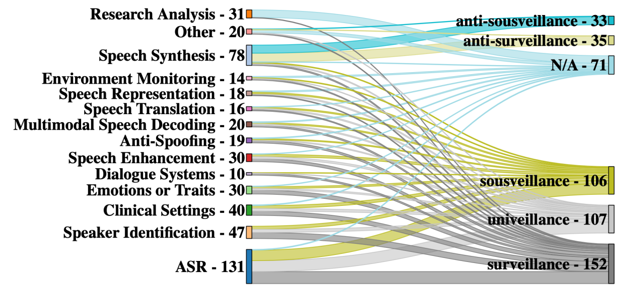

Figure 2: Sankey diagram showing the relationship between task categories and axes (including N/A) from Veillance Compass. Papers may belong to multiple task categories and axes; link widths indicate the number of papers per category-axis pair.

as these only arose as a necessity for analysis and therefore have not been subject to multiple annotations. For the veillance compass axes, we compute agreement for each axis individually.

We find substantial agreement between the two annotators, with perfect agreement around discussions of dual use and surveillance. The lowest agreement scores are for sousveillance and univeillance; however in discussion of annotation differences, we found that the difference was caused because one annotator did not label the least afforded veillance, if they perceived a difference in their affordances, whereas the other annotated all types of veillances that were afforded, regardless of differences in how much they were afforded. For example, in the disagreement regarding whether a paper producing a model for *Speech enhancement* in noisy conditions should be tagged as univeillance, one annotator's position was that it affords all kinds of veillances equally, while the other annotator argued that it is less likely for one-to-one setting to be noisy, thus it affords surveillance and sousveillance more than univeillance. The annotators chose to not change their respective approaches in this case, and considered it as an interesting case, that have a potential to be explored in further possible applications of the theory of affordances for surveillance carried by speech processing technology.

Annotator Information The annotators involved in this work are both authors of this paper, and were not compensated for their work. They are both in their 30s and have received formal training in computer science, and are currently working in research positions.

4 Results

4.1 Tasks and Veillances

From our analysis of the relationship between tasks and veillances (see Figure 2), we find that only 2 papers ($< 1\%$) in our sample mention surveillance at all, both of which merely mention surveillance, while 56% of all papers are labelled with `surveillance`. Of the two papers that do highlight surveillance, they only do so in passing, with reference to how their work affords surveillance. For example, Liang et al. (2025) note that:

> [lipreading] also plays an important role in silent secret communication scenarios and long-distance surveillance.

Calling to attention the negative potential of their work while Tao, Jia, and Hu (2025) call attention to

> Accurate knowledge of source positions contributes to a range of practical audio-only applications, such as robot navigation, cochlear implants and intelligent surveillance systems

highlighting the potential for surveillance systems as a positive.

The discrepancy of proportion of papers that mention surveillance and the proportion tagged with `surveillance` is even more stark when we consider that we map 170 papers (74%) of the sample to one or more axes of the veillance compass. Considering Figure 2, we can see how tasks distribute to the different the veillance axes: *Environment Monitoring*, *Speech Representation*, *Speech Translation*, *Multimodal Speech Decoding*, *Anti-Spoofing*, *Speech Enhancement*, *Dialogue Systems*, *Emotions or Traits*, *Clinical Settings*, *Speaker Identification*, and *ASR* all exclusively contribute to sur-, sous-, and univeillance of the compass axes. Of these, ASR is the largest contributor, contributing with 50 out of 53 papers on ASR to sur-, sous-, and univeillance. Two papers that are labelled as N/A for surveillance are focused on different kinds of ASR analysis (Serditova, Tang, and Steffens 2025; Pratap Singh, Sahidullah, and Kinnunen 2025). The third N/A focuses on voice conversion-resistant watermarking. Therefore, these papers do not produce any models or data, which does not allow for them to contribute towards any type of veillance. In fact, only the *Speech synthesis* and *Other* categories contribute towards anti-surveillance and anti-sousveillance.

Within the categories of anti-surveillance and anti-sousveillance, the majority of papers describe efforts towards various forms of speech synthesis, tasks such as voice conversion, text-to-speech (TTS), and spoofing. These tasks all share the feature that they afford concealing identifiable markers of ones own voice, while still producing speech. Famously, the activist-hacker collective "Anonymous" have used speech synthesis systems to avoid detection and arrest. Speech synthesis tasks have recently come under greater scrutiny for the risks of social harms associated with them, highlighting the importance of taking a nuanced approach to discussing the benefits and harms of different areas of work in speech technology research.

Task-based Analysis As we have noted above, not all tasks give rise to the same set of risks. Here, we provide a fine-grained analysis of what each task category gives rise to.

ASR: An important component of surveillance is not just recording, but also the ease with which monitoring can occur. ASR affords making speech more easily searchable and thereby afford easier monitoring and indexing of the contents of speech and thus affords surveillance, sousveillance and univeillance.

Speech Translation: Where ASR provides easier searchability, speech translation affords cross-lingual monitoring of the contents of speech.

Speaker Identification: Speaker identification tasks affords tracking individuals, and disambiguating their speech from others.

Speech Enhancement: Where speech is distorted by noise or otherwise poor recording quality, speech enhancement can allow for recovering speech and thereby can afford surveillaince, sousveillance, and univeillaince when combined with tools such as ASR and Speech Translation.

Speech Synthesis: Through mechanisms such as voice conversion, tasks within speech synthesis affords mechanisms for individuals to mask their own voice, and the biological markers that it communicates, and thereby allows for people to protect their identity and limit identity-based tracking. Yet synthesis does not, by design, prevent risks of identification or tracking of the contents of speech.

Anti-Spoofing: In the goal of its task, anti-spoofing allows for identifying (segments of) audio which has been synthesised, thereby mitigating the benefits of speech synthesis, and can in this way afford reconstructing surveillance, sousveillance, and univeillance of content that a speaker may have synthesised for protecting their identities.

Emotions or Traits (paralinguistics): Tasks within paralinguistics seek to identify information about the speaker, such as their emotional states, accents, warmth, and so forth. Through such efforts, paralinguistics offers a mechanism through which one can go beyond identifying the speaker or the content of speech, to also infer personal characteristics and sensitive attributes of a speaker.

Clinical Settings: Speech technologies developed for clinical settings afford the identification of sensitive medical information such as health disorders, disabilities, and neurological conditions.

Multimodal Decoding: In our analysis, multimodal decoding covers a variety of tasks such as sentence prediction using EEG and producing speech from lipreading. Collectively, the risks of surveillance, sousveillance, and univeillance that arise from this set of tasks is producing speech from non-audio signals, thereby allowing for reconstruction of communication that a given speaker may have been explicitly trying to keep private.

Speech Representations: While developing better speech representations constitutes basic research within modelling speech signals, it also affords improvements on downstream tasks, including ASR, and can therefore aid across tasks in terms of anti-surveillance and anti-sousveillance as well as their counterparts.

Dialogue Systems: Unlike many other types of systems, dialogue systems allow for many similar mechanisms of surveillance; however, dialogue systems are designed to be opt-in systems, thereby allowing the speaker control over whether they will engage with them. However, in many cases, such as customer service, individuals may only be able to access engaging with services through dialogues sys-

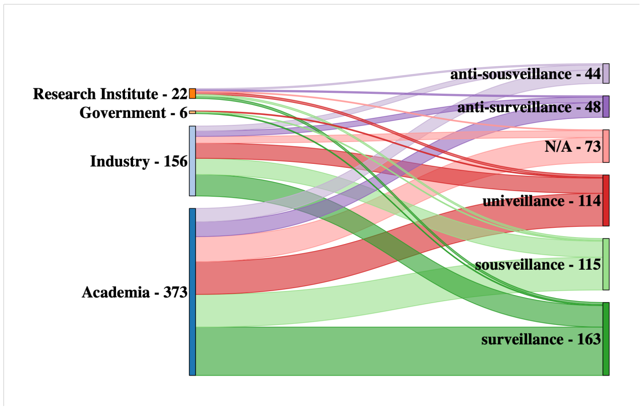

Figure 3: Relationship between author affiliations and axes (including N/A) from Veillance Compass.

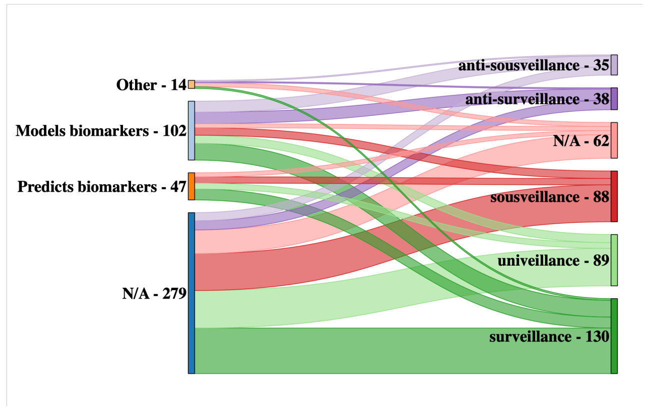

Figure 4: Relationship between biomarker usage types and axes (including N/A) from Veillance Compass.

tems. Yet, even within those instances, they are inherently more limited in that they require the active participation of the speaker.

Environment Monitoring: Environment monitoring does not directly engage with speech. However, such systems afford collecting information about the physical surroundings and contexts within which a speaker is situated at the moment of recording and thereby affords surveillance and univeillance not of the speech, but rather of the speaker and their immediate context.

Research Analyses & Other: Research analyses and the other category cover many different areas. The shared factor for them, however is that they are not readily used for either monitoring or resisting being monitored. The exception to this rule is the development of analytical datasets and methods for assessing fairness (e.g., Sakuma et al. 2025), which provides ethnicity, nationality, and race, which can allow the developers of systems to identify the populations for which their tools fail, and thus aid in enrolling users into surveillance infrastructures that they would otherwise have evaded.

4.2 Dual Use, Risks, and Limitations

In our findings, we that the vast majority of papers do not include discussions around dual use, risks that arise from the work, or even the limitations of the work itself. Indeed, only 1.3%, 5%, and 25% of papers discuss or mention these, respectively. While dual use has not been common to report in many computational fields, we follow the recommendations set out by Kaffee et al. (2023), for natural language processing. The questions presented there around the processes, development, and description of research artefacts will also be useful for the speech technology research field.

One example of engaging with the question of risk is provided by Kondo et al. (2025), where they remind that

> Users must be mindful that the idols are actively involved in the entertainment industry, and care must be taken to avoid harming the reputation of the recorded groups, their management companies, fans, or any associated parties

when using their speech corpus of Japanese idol speakers.

4.3 Affiliations, languages, and artefact availability

While governmental agencies and research institutes occur too infrequently in our sample to draw any conclusions, we find an interesting pattern among academia and industry (see Figure 3). Although academia has primarily focused on sur-, sous-, and univeillance ($\sim$ 56%), approximately 17% of all academic work focuses on anti-surveillance and antisousveillance. While industry papers have a similar proportion ($\sim$ 16%), this is a shadow number of work that occurs in industry. Kalluri et al. (2025) examine patents related to computer vision technologies and find that there is a research to surveillance pipeline in industry. That is, while there is no reason that academic–and public industry–work would not align along the identified patterns, only a portion of private industry work is made public and propriety work tends toward veillance products, often for purposes such as profiling and targeted marketing (Iqbal et al. 2022).

Unsurprisingly, the most prevalent language of study is English, covered by approximately half of all papers. Following which are Chinese languages (Mandarin, Cantonese) and dialects, accounting for 12% of papers, and then 10% of all papers were on multilingual speech processing, covering a wide array of languages. Although one should expect that languages that are covered in a line of work to be clearly stated, we find that it was often necessary to consult the datasets themselves to identify which languages were included.

In terms of resource availability, we find that 74 papers report that their data, code, or methods are available, while only one paper limits access. The remaining papers either are papers for which there are no research artefacts of note to share (20 papers) , or the authors have not explicated how to access their research artefacts (133 papers).

4.4 Biomarkers

We find that 33.2% of papers include modelling or predicting biomarkers spanning across all veillance axes. While biomarker prediction seems to be used only in veillance technologies when it is applicable, 37% of works that modelled biomarkers contribute to counterveillance (see Figure 4 for visuals), which is expected, as modelling specific voice characteristics is crucial for realistic voice synthesis.

5 Discussion

So far in this paper, we have outlined Mann's veillance compass, developed and analysed a dataset of Interspeech papers, and presented our results. Here, we shift to questions and insights that have arisen from the papers, our discussions, and the literature in the field.

5.1 Does a tool afford the same for everyone?

Both annotators hold degrees in computer science, so they would be able to implement and run the tools, even where code is not given is for the papers presented, given the necessary amount of compute and appropriate data. However, much of the work that is analysed requires significant background knowledge of programming and computing, and would therefore not be as easily affordable to a lay person.

In a similar way, where a technology requires large-scale compute to be able to run, it may, in theory, afford the ability to perform ASR across dozens of languages, but will only be available to a small handful of entities that have a large enough budget for computing. In this case, the technology itself affords surveillance, and to a lesser degree sousveillance. When a technology, on the other hand, requires little compute or expertise to deploy and use, it also affords being used by a much wider base, and can thus more easily afford sousveillance, even as it also offers surveillance capabilities for an entity with a large compute budget.

5.2 What are the implications of a tool affording surveillance?

There are several implications of a tool that affords surveillance depending on the context. For example, work on language technologies for low-resource languages has the potential of aiding people in meeting their needs in new ways that are easier to access (Yan et al. 2025), however, the creation of language resources and technologies can also subject a population that had not been under much surveillance to greater degrees of surveillance.

Yet, simply because a tool holds the potential for surveillance does not mean that we should not work on it. Instead, it is important that we develop our research infrastructures, with the potentials risks of it in mind and with an eye to formulating methods to address and mitigate those risks. That is, rather than seeking sanctions for particular lines of work, we should seek to develop infrastructures that can allow for research to be conducted and research artefacts produced without negative impacts to speaker communities.

5.3 Developing Countermeasures

While on one hand developing countermeasures against surveillance by design can aid people in avoiding being held accountable for illegal activities, for example, it can also help a wider community engage in modern life with greater degrees of privacy. It is therefore useful to dedicate resources towards the development of tools and mechanisms that can help mitigate the risks of being under surveillance, while at the same time engaging in care with how such tools should be released, and in both cases release of tools for and against surveillance should be handled with care.

While research in NLP has extensively addressed questions of social harms of technologies (e.g., Kaffee et al. 2023; Bender and Friedman 2018), such methods are not readily portable to speech research due to the additional information conveyed through human speech. For example, unlike text, speech communicates biometric and paralinguistic information in addition to linguistic information, which makes it possible to track speakers, identify them, and evaluate specific biomarkers (Kröger, Lutz, and Raschke 2019). We therefore argue that it is necessary to develop approaches that take into account the wide variety of personal surveillance risk that speech technologies pose.

5.4 Contemporary Research Practices

We find it greatly concerning that research papers almost entirely omit the limitations of their studies. While it has negative implications for the research community more widely, in that research is not properly hedged, it also has negative implications for identifying potential mechanisms for mitigating the risks of surveillance that are posed by research tools, and can stifle research into the development of methods for preserving privacy.

Relatedly, we take at face value what is described in papers, and do not systematically validate whether artefacts indeed are available at their described level. It is therefore possible that research that claims to be open-source has not actually been released or can be accessed without directly contacting the authors.[4]

A surprising and hope-inducing finding from C M et al. (2025), who experiment with, and find that ASR performance decreases when microphones are capturing speech through a speaker. We find this work particularly desirable direction for thinking about interventions that may help individuals circumvent surveillance mechanisms, particularly if they are participating in pro-democratic activities such as protests.

6 Recommendations

Drawing on the results of our annotation and the following discussion, we propose our recommendations for conducting research in speech processing with consideration of potential risks tied to surveillance. Additionally, we provide a list of questions that might be used for evaluating such risks. Firstly, languages on which research is conducted should be explicitly mentioned (Bender 2019). Secondly, due to the risks of malicious use of research outputs, research artefacts should clearly specify permissible and impermissible uses, e.g. through licensing. Thirdly, we recommend that conferences provide space for submitting links to research artefacts when submitting camera-ready versions and develop

[4]Indeed, in the small set of links provided in papers that we followed—to validate whether merely samples were made public or code and/or data—some were empty webpages or broken links.

a repository for research outputs with access controls and policies to balance transparency and risk mitigation.

6.1 Incentives for developing tools

In the development of tools that afford surveillance or mitigate it, we should critically engage with the incentives of different stakeholders and how they might use such tools. To examine how a tool might be used, researchers can ask:

- What does our work afford in terms of surveillance or mitigation thereof?
- Which stakeholders might wish to make use of our tool and to what ends?
- What are mechanisms by which we can ensure that our intended use of the tools aligns with the actualised use of them?

Different stakeholders may also have different motivations for working on tasks. For example, industry researchers may have less agency to work on measures that mitigate risks of surveillance due to the potential risks to their public image, should their tool be used for negative or harmful outcomes.

6.2 Field-level recommendations

We recommend that venues in the speech research community allow for sections that are not counted towards the page limit. While it is disappointing that concerns like dual use, limitations, and surveillance are not further discussed in papers, we believe that this issue is systemic. Given limited space, researchers are likely to dedicate as much space as possible towards describing their research in detail. By allowing for sections such as limitations that do not count towards the page count, the field can encourage researchers to critically engage with their work, which we believe will have outsized benefits for the field as a whole.

6.3 Paper-level Recommendations

For researchers writing such sections, we recommend that they ask and attempt to answer the following questions, which we believe would be a valuable starting point for opening a conversation about the relationship between speech research and surveillance.

- How might our work be used to improve surveillance capabilities of speech technologies?
- What are (social or technological) mechanisms to mitigate the surveillance impacts of your work?
- Whose speech does the research outcome target?
  - If the group is linguistically underrepresented, have they accepted for tools for their language to being publicly shared? If not, how are you distributing the methods? If yes, then under what conditions, if any?
  - If the group is vulnerable (for example, children, elderly people, or individuals with mental or physical disabilities affecting their cognitive abilities), what protections have been implemented to protect the people who may be targeted using this work?
- What outcomes does your research afford?
- Is your research easy to implement and use (in case of model)?
- How much computational power does your model need for training and inference?
- How much data was used to develop your model?
- What are the limitations of your research?
- What open questions remain or have arisen from your work?

By addressing these questions, researchers will be better suited to answer and address the question of surveillance and social harms of speech technologies, and thereby, hopefully, identify pathways towards more socially just and desirable speech technologies.

6.4 Potential Research Directions

Our final set of recommendations pertains to open research directions. While there is a great deal of research on how to build better and more impressive models for a variety of tasks, we found very limited evidence of research that seeks to engage with **developing privacy-preserving methods** or with developing methods that can disturb the performance of methods that afford surveillance. Moreover, there is space in the speech field for thinking about how to distribute and share data and models that maximise their social benefit, i.e., allow for the tool to exist without its surveillance affordances being realised. We believe that attending to these research areas can provide safer and more inclusive outcomes for the research field as a whole.

7 Limitations

This study has several limitations. First, the work has no longitudinal validity, as it examines one specific year in one specific venue. Second, we discovered that some papers labelled as open-source may not actually be publicly available online. Third, while our sample is 20% of all Interspeech 2025 papers, certain affiliation types, particularly governmental agencies and research institutes, appear very infrequently in our sample, and we therefore cannot draw strong conclusions on the results of such affiliations, or the tasks that they tend to work on. Additionally, although we are confident that our annotation of languages covered is correct, there may be instances where newer versions of the datasets that include more languages have been used.

8 Conclusion

In this paper, we highlight the importance of the relationship between speech technologies and surveillance by providing the historical background of surveillance and explaining how speech processing contributes to the rise of mass surveillance. We then analyse randomly selected 20% sample of papers from Interspeech 2025. The results confirm that research on speech processing can afford veillances and demonstrate how rarely researchers engage with the related risks. Based on these findings, we provide recommendations and questions for researchers to consider in their future work, encouraging awareness of the risk of how their work can be used for surveillance.

9 Generative AI Disclosure

We have not used generative AI, or any other form of AI, for any step in this research project.

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